

TECHBIG SOLUTIONS

SAP S/4HANA – Extended Warehouse Management (EWM)

Complete Professional Training Syllabus

Ambattur, Chennai

Course Objective

A comprehensive functional and implementation-oriented syllabus for SAP S/4HANA EWM covering embedded and decentralized EWM concepts, warehouse structure, master data, inbound, outbound, internal warehouse operations, warehouse tasks and orders, storage control, RF, physical inventory, labor/resource optimization, automation, monitoring, integration, configuration and production support.

The curriculum is aligned with SAP's current EWM learning areas: warehouse structures and master data, inbound/outbound processing, storage control, physical inventory, warehouse monitoring/mobile devices, resource optimization and advanced warehouse processes. ■cite■turn0search2■turn0search5■

1. SAP S/4HANA EWM Overview

- What is SAP EWM and why it is used
- EWM versus Inventory Management and classic LE-WM
- Embedded EWM versus decentralized EWM
- Basic versus advanced warehouse in supported S/4HANA scenarios
- EWM architecture and system landscape
- End-to-end warehouse process overview
- SAP Fiori and warehouse-user experience
- EWM integration with S/4HANA and other SAP solutions

2. Enterprise & Warehouse Organizational Structure

- Plant and storage location relationship
- Warehouse number
- Storage type
- Storage section
- Storage bin
- Activity area
- Work center
- Door and staging area
- Warehouse process areas
- Physical warehouse layout to EWM structure mapping

3. EWM Master Data

- Product master and warehouse-product data
- Warehouse-specific product attributes
- Packaging specifications
- Handling unit concepts
- Packaging materials
- Units of measure
- Batch and serial number management
- Warehouse task-relevant master data
- Resource and labor-related master data
- Master-data validation and troubleshooting

4. Warehouse Process Types & Determination

- Warehouse process type concept
- Process categories
- Warehouse process type determination
- Putaway process types
- Picking process types

- Internal movement process types
- Posting change process types
- Warehouse task creation controls
- Warehouse order creation controls
- Exception handling

5. Inbound Processing – Complete Cycle

- Inbound delivery integration
- Purchase order to inbound delivery
- Transportation unit and vehicle overview
- Arrival and unloading
- Goods receipt
- Deconsolidation
- Quality inspection integration
- Putaway warehouse tasks
- Putaway confirmation
- Handling units in inbound
- Exception handling and reversals

6. Putaway Strategies

- Fixed bin
- Empty bin
- Addition to existing stock
- Near picking area
- Pallet storage and bulk concepts
- Capacity-based determination
- Section-based determination
- Warehouse process type and search sequence
- Storage type search
- Putaway with handling units
- Troubleshooting bin determination

7. Outbound Processing – Complete Cycle

- Sales order to outbound delivery
- Outbound delivery order
- Warehouse task creation
- Picking
- Pick denial and short pick
- Packing
- Staging
- Loading

- Goods issue
- Handling units and shipment preparation
- Returns and reversal scenarios

8. Stock Removal / Picking Strategies

- FIFO and shelf-life-related concepts
- Fixed bin picking
- Stock removal control
- Batch and serial selection
- Quantity-based removal
- Storage type search
- Consolidation concepts
- Pick denial and exception codes
- Wave-related picking scenarios
- Troubleshooting source-bin determination

9. Warehouse Tasks (WT)

- Warehouse task concept
- Create, confirm and cancel WT
- Source and destination data
- Warehouse task categories
- Warehouse task confirmation
- Partial confirmation
- Difference handling
- Exception codes
- Task monitoring
- Mass processing

10. Warehouse Orders (WO)

- Warehouse order concept
- Grouping warehouse tasks
- Warehouse order creation rules
- Sorting and queue assignment
- Priority management
- WO confirmation and monitoring
- Resource assignment
- Labor/resource optimization considerations

11. Storage Control

- Layout-oriented storage control
- Process-oriented storage control

- Multi-step inbound process
- Deconsolidation and repacking
- Intermediate storage
- Complex outbound flows
- Warehouse task sequencing
- Integration with work centers
- Exception handling in storage control

12. Handling Unit Management

- Handling unit structure
- HU creation and packing
- Nested handling units
- Packaging specifications
- HU movement
- HU-based putaway and picking
- HU labels
- HU split/merge
- HU stock visibility
- HU integration with delivery processes

13. Internal Warehouse Processes

- Ad hoc warehouse movements
- Bin-to-bin transfer
- Posting changes
- Stock transfers
- Rearrangement
- Replenishment
- Automatic, planned and order-based replenishment
- Warehouse-internal movements
- Stock corrections
- Exception processing

14. Physical Inventory

- EWM physical inventory concepts
- Inventory procedures
- Create physical inventory documents
- Count storage bins
- Enter counts
- Recount
- Difference analysis
- Difference posting

- Cycle counting
- Continuous inventory
- Troubleshooting inventory differences

15. Replenishment & Rearrangement

- Fixed-bin replenishment
- Automatic replenishment
- Planned replenishment
- Order-based replenishment
- Minimum/maximum quantities
- Rearrangement of stock
- Fast-moving product optimization
- Monitoring replenishment tasks

16. Slotting & ABC Analysis

- Purpose of slotting
- Warehouse product data for slotting
- ABC classification
- Putaway optimization
- Storage section and bin determination
- Capacity and product characteristics
- Re-slotting concepts
- Performance and operational benefits

17. Cross-Docking

- Cross-docking concept
- Planned cross-docking
- Opportunistic cross-docking overview
- Inbound-to-outbound flow
- Avoiding unnecessary putaway
- Cross-docking determination
- Operational monitoring

18. Value-Added Services (VAS)

- VAS concepts
- Repacking and labeling
- Kitting
- De-kitting / reverse kitting
- VAS orders and warehouse tasks
- Work-center processing
- Customer-specific warehouse services

19. Yard Management & Shipping/Receiving

- Yard and door concepts
- Transportation units
- Vehicles
- Check-in/check-out overview
- Door determination
- Staging and loading
- Shipping/receiving processes
- Integration considerations

20. RF Framework & Mobile Warehouse Operations

- RF framework overview
- RF devices and users
- RF menu structure
- RF picking
- RF putaway
- RF internal movements
- Barcode scanning
- Exception processing on RF
- RF troubleshooting

21. Resource Management & Labor Optimization

- Warehouse resources
- Resource types
- Queues
- Resource assignment
- Resource groups
- Activity areas
- Warehouse workload
- Labor management overview
- Performance monitoring
- Resource optimization

22. Automation & Material Flow System (MFS)

- Warehouse automation overview
- Conveyor and automated storage concepts
- MFS architecture
- Communication points
- Warehouse control units
- Telegram concepts

- Automated warehouse processes
- Monitoring and troubleshooting
- Integration with automated equipment

23. Advanced EWM Functions

- Wave management
- Wave templates
- Wave release
- Pick-pack-pass concepts
- Cartonization overview
- Kit-to-stock and kit-to-order
- Deconsolidation
- Complex internal routing
- Transit warehousing overview
- Advanced warehouse analytics

24. EWM Integration with S/4HANA

- MM integration
- SD integration
- PP integration
- QM integration
- TM integration overview
- Production staging
- Goods movements
- Delivery integration
- Batch and serial integration
- Accounting and inventory reconciliation

25. EWM Monitoring & Analytics

- Warehouse Monitor
- Monitor nodes and document monitoring
- Warehouse stock and bin monitoring
- Open warehouse tasks
- Warehouse orders
- Resource monitoring
- Workload monitoring
- Exception monitoring
- KPI dashboards
- CDS-based analytics overview
- Fiori analytical apps

26. EWM Configuration / SPRO Curriculum

- Define warehouse number and assignments
- Configure storage types, sections and bins
- Define activity areas
- Warehouse process types
- Warehouse task determination
- Warehouse order creation rules
- Storage type search
- Putaway strategies
- Stock removal strategies
- Packaging specifications
- Handling unit settings
- Storage control
- Wave management
- Physical inventory
- Replenishment
- RF framework
- Resource/queue configuration
- PPF and action determination overview
- Integration configuration

27. Post Processing Framework (PPF)

- PPF purpose
- Action profiles
- Condition technique
- Automatic actions
- Document-triggered processing
- Inbound/outbound actions
- Monitoring actions
- Troubleshooting failed actions

28. Migration & Implementation Methodology

- EWM implementation phases
- Fit-gap analysis
- Warehouse blueprint
- Master-data migration
- Initial stock considerations
- Cutover planning
- Integration testing
- Unit/system/integration/UAT testing

- Go-live preparation
- Hypercare and support
- Migration from legacy WM concepts

29. Production Support & Troubleshooting

- Inbound delivery not creating warehouse documents
- Outbound delivery and picking issues
- WT/WO creation failures
- Incorrect storage type/bin determination
- Stock inconsistency
- HU issues
- Batch/serial issues
- Queue/resource issues
- RF errors
- PPF/action failures
- Physical inventory differences
- Integration errors
- Root-cause analysis and incident documentation

30. Real-Time End-to-End Projects

- Project 1: PO → inbound delivery → GR → putaway
- Project 2: Sales order → outbound delivery → wave → picking → packing → PGI
- Project 3: HU-based warehouse process
- Project 4: Replenishment of fixed picking area
- Project 5: Physical inventory and difference processing
- Project 6: Cross-docking scenario
- Project 7: VAS/kitting scenario
- Project 8: RF-enabled warehouse execution
- Project 9: Automated warehouse/MFS overview
- Project 10: Full implementation + production-support case

31. Interview & Certification Preparation

- EWM configuration questions
- Inbound/outbound scenario questions
- Storage control scenarios
- WT/WO questions
- RF and automation questions
- Integration questions
- Real-time support cases
- Implementation methodology
- Mock interviews

- Practical assessment

Recommended Training Roadmap

Phase	Coverage	Outcome
1	Foundation & Structure	Understand EWM architecture and warehouse structure
2	Master Data & Configuration	Build warehouse master data and core configuration
3	Inbound & Putaway	Execute end-to-end receiving and putaway
4	Outbound & Picking	Execute wave, picking, packing, staging and PGI
5	Internal Operations	Replenishment, rearrangement, posting changes and PI
6	Advanced EWM	RF, VAS, cross-docking, slotting, resources and automation
7	Integration & Support	Integrate S/4HANA modules and troubleshoot
8	Project	Complete end-to-end implementation and support case

SAP Reference Alignment

SAP's current EWM learning content covers warehouse structures and master data, inbound/outbound goods movements, storage control, physical inventory, warehouse monitoring, mobile devices, resource optimization and advanced warehouse processes. SAP also identifies EWM as the strategic warehouse-management application for S/4HANA and supports embedded or decentralized deployment depending on the scenario and edition.

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